

MUTUAL FUND ANALYTICS

BlueStock Fintech — Capstone Project I

Final Project Report

End-to-end ETL, EDA, Performance Analytics, Risk Analysis and
Interactive Dashboard

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Executive Summary

This project develops an end-to-end Mutual Fund Analytics solution using fund master data, daily NAV history, fund-house AUM, SIP activity, category inflows, investor transactions, scheme performance, portfolio holdings and benchmark indices. The objective is to convert heterogeneous financial datasets into reliable analytical outputs and an interactive business dashboard.

The workflow combines Python-based ingestion and cleaning, SQLite/SQL analysis, exploratory data analysis, performance metrics, historical risk analysis and Power BI visualization. The core NAV universe contains 40 mutual fund schemes, while the investor transaction dataset contains 32,778 records and the portfolio holdings dataset contains 322 holding records.

The analysis shows strong growth in SIP activity and folios over the available period. Monthly SIP inflow reached approximately ■31,002 crore in December 2025, while industry folios reached 26.12 crore. Performance analysis demonstrates why risk-adjusted measures such as Sharpe ratio, volatility, drawdown and VaR are necessary alongside headline returns.

Advanced analytics extend the project into investor cohorts, SIP continuity, a simple risk-based recommender and portfolio concentration using the Herfindahl-Hirschman Index (HHI).

Overall conclusion: mutual-fund decision support is strongest when industry trends, investor behavior, fund performance, downside risk and portfolio concentration are evaluated together.

1. Project Background and Objectives

1.1 Background

Mutual fund platforms generate multiple forms of structured and time-series data. Fund metadata describes schemes and risk characteristics, NAV data captures daily market-linked movement, SIP datasets show recurring investment behavior, and investor transactions provide demographic and geographic context. A unified analytics workflow can transform these sources into decision-support information.

1.2 Objectives

- Build a reproducible data ingestion and cleaning workflow.
- Create a structured SQLite database and SQL query layer.
- Perform exploratory analysis of NAV, AUM, SIP, investor and portfolio data.
- Evaluate returns and risk-adjusted performance across schemes.
- Calculate Historical 95% VaR and CVaR for 40 schemes.
- Analyze investor cohorts and SIP continuity.
- Build a simple risk-based fund recommender.
- Measure portfolio concentration using HHI.
- Deliver an interactive Power BI dashboard and professional project outputs.

2. Data Sources and Data Dictionary

Dataset	Purpose	Key fields
Fund Master	Scheme metadata	amfi_code, category, benchmark, expense_ratio, risk_category
NAV History	Daily scheme valuation	amfi_code, date, nav
AUM by Fund House	Industry/fund-house scale	fund_house, date, aum_crore, aum_lakh_crore
Monthly SIP Inflows	SIP activity	month, sip_inflow_crore, active_sip_accounts
Category Inflows	Category-level flows	category, month, net_inflow_crore
Industry Folios	Investor account scale	month, total_folios_cr
Scheme Performance	Performance/risk metrics	returns, alpha, beta, Sharpe, drawdown
Investor Transactions	Investor behavior	investor_id, date, amount, state, age_group
Portfolio Holdings	Security/sector exposure	stock, sector, weight_pct, market_value_cr
Benchmark Indices	Market comparison	index name, date, close value

The datasets were standardized around common identifiers such as AMFI scheme codes and dates. Fund-level tables were linked where an AMFI code was available, while aggregate industry datasets were analyzed independently to avoid creating artificial relationships.

3. ETL Design and Data Preparation

3.1 ETL workflow

The ETL design follows a Raw → Clean → Analyze → Report pattern. Raw CSV files are preserved separately from processed outputs. Python scripts handle ingestion, validation and cleaning before analytical notebooks and Power BI consume the prepared datasets.

- Load raw CSV datasets with Pandas.
- Parse dates into consistent datetime types.
- Standardize column names and numeric fields.
- Sort NAV observations by scheme and date.
- Calculate daily returns using percentage change.
- Validate AMFI identifiers and missing values.
- Create processed analytical datasets.
- Load relational data into SQLite for structured querying.

3.2 Data-quality considerations

Time-series analysis requires careful treatment of dates. Trading data does not occur on every calendar day, so analysis should respect available market observations and avoid treating weekends as missing investment information. Financial calculations are also sensitive to annualization conventions and should use trading-day assumptions where appropriate.

The repository separates raw and processed data, keeps database schema definitions in SQL, and excludes the SQLite database file from Git tracking.

4. Exploratory Data Analysis — Industry Trends

4.1 AUM

Fund-house AUM was analyzed across the available monthly history. The Power BI industry page presents the latest AUM, fund-house comparison and the industry AUM trend. The latest dataset-derived AUM value displayed in the dashboard is 54.47 lakh crore.

4.2 SIP inflows

SIP activity is one of the strongest indicators of recurring investor participation. Monthly SIP inflow reached approximately ■31,002 crore in December 2025. Active SIP accounts and new SIP accounts were also analyzed to distinguish recurring participation from new account formation.

4.3 Folios

Industry folios increased from approximately 13.26 crore in January 2022 to 26.12 crore in December 2025. This represents a substantial expansion in the number of mutual-fund accounts represented in the dataset.

Interpretation: the combination of increasing folios and strong SIP inflows indicates broadening participation in systematic investing, although the dataset does not by itself establish causality.

5. Exploratory Data Analysis — Investor Behavior

5.1 Demographic analysis

Investor transactions were analyzed by age group, gender, state, city and city tier. The 26–35 age group represents the largest investor segment in the analyzed transaction dataset.

5.2 Transaction activity

The investor dataset contains 32,778 transaction records. Transaction amount and transaction type were visualized in the Power BI Investor Analytics page. State-level analysis highlights geographic differences in transaction activity.

5.3 SIP behavior

A dedicated SIP subset contains 19,716 SIP transactions across 4,762 unique SIP investors. This subset was used for continuity analysis rather than treating all investor transactions as SIP activity.

5.4 Portfolio holdings

The portfolio holdings dataset contains 322 records across 34 funds. Sector allocation was aggregated to identify exposure patterns and concentration differences.

6. Exploratory Data Analysis — Visual Findings

The EDA package contains 16 exported visualizations covering the major dimensions of the project.

Area	Visual outputs
NAV	EDA_NAV_Trends.png; EDA_NAV_Trends_2023_2024.png
AUM & industry	EDA_AUM_Growth.png; EDA_Folio_Growth.png
SIP	EDA_SIP_Inflows.png; EDA_Active_SIP_Accounts.png; EDA_SIP_AUM_Growth.png; EDA_New_SIP_Accounts.png
Categories	EDA_Category_Inflows_Heatmap.png
Investor demographics	EDA_Age_Distribution.png; EDA_Gender_Distribution.png; EDA_SIP_Amount_by_Age.png
Geography	EDA_SIP_by_State.png; EDA_T30_vs_B30.png
Performance	EDA_Return_Correlation.png
Portfolio	EDA_Sector_Allocation.png

Together, these visuals establish the descriptive foundation for the later performance, risk and advanced analytics sections.

7. Performance Analysis

7.1 Return metrics

Scheme performance was evaluated using 1-year, 3-year and 5-year returns. Benchmark comparison was incorporated where benchmark return fields were available. The objective is to compare absolute performance while retaining context about the market or benchmark.

7.2 Risk-adjusted metrics

The analysis includes Alpha, Beta, Sharpe ratio, Sortino ratio, annualized standard deviation and maximum drawdown. Sharpe ratio relates excess return to volatility, while Sortino focuses on downside variability. Beta captures sensitivity to a benchmark, and Alpha represents performance beyond the benchmark relationship under the selected model.

7.3 Supporting outputs

- alpha_beta.csv
- fund_scorecard.csv
- tracking_error.csv
- Performance_Analytics.ipynb

These outputs support both scheme-level comparison and the Power BI Fund Performance page.

8. Risk Analytics — VaR and CVaR

8.1 Historical VaR

Historical 95% VaR was calculated from daily NAV returns for each of the 40 schemes. The 5th percentile of the daily return distribution represents the historical downside threshold at the selected confidence level.

8.2 CVaR

CVaR, also called Expected Shortfall, was calculated as the mean of returns falling at or below the 5th percentile threshold. It therefore describes the average loss severity in the historical tail rather than only the threshold.

8.3 Interpretation

More negative VaR or CVaR values indicate greater historical downside risk. These metrics should not be interpreted as guarantees of future losses; they summarize the observed historical distribution.

```
Output: data/processed/var_cvar_report.csv
```

9. Rolling Sharpe Analysis

A 90-day rolling Sharpe ratio was calculated for five key funds using daily NAV returns. The rolling approach provides a time-varying view of risk-adjusted performance rather than a single full-period score.

Selected fund	Purpose
HDFC Top 100 Fund	Large-cap equity reference
HDFC Short Term Debt Fund	Debt-oriented comparison
HDFC Mid-Cap Opportunities Fund	Mid-cap equity reference
ABSL Frontline Equity Fund	Large-cap equity reference
ABSL Small Cap Fund	Small-cap equity reference

The chart is saved as `rolling_sharpe_chart.png`. Comparing rolling values helps identify periods when a fund's return profile became more or less attractive relative to its recent volatility.

10. Investor Cohort Analysis

Investors were grouped by the year of their first transaction. This creates a cohort structure that allows investor groups entering in different years to be compared.

Measure	Description
Cohort year	Year of first recorded transaction
Investors	Unique investors in the cohort
Total invested	Sum of transaction amounts
Average transaction	Mean transaction amount
Top fund	Most frequently transacted scheme by cohort

The output is stored in `investor_cohort_analysis.csv`. Cohort analysis can help identify changes in investor scale and preferences across entry periods.

Limitation: the average amount measure is based on transaction amounts in the analyzed cohort rather than being a pure SIP-only measure.

11. SIP Continuity Analysis

SIP continuity was analyzed using only transactions classified as SIP. Investors with at least six SIP transactions were treated as eligible for gap analysis.

Measure	Result
SIP transactions	19,716
Unique SIP investors	4,762
Eligible investors	1,362
At-Risk	1,332 (97.80%)
Regular	30 (2.20%)

The status rule classified investors with an average SIP transaction gap above 35 days as At-Risk and the remainder as Regular.

The high At-Risk percentage should be interpreted cautiously. It is a rule-based analytical classification based on observed transaction gaps, not evidence that these investors have actually stopped their SIPs.

12. Fund Recommendation System

A lightweight recommender was implemented using scheme risk grades and Sharpe ratio. Original risk grades were mapped into Low, Moderate and High groups. Within each group, schemes were ranked by Sharpe ratio and the top three were returned.

Risk group	Example top-ranked schemes	Sharpe
Moderate	HDFC Top 100 Fund; Mirae Asset Large Cap Fund; ICICI Pru Bluechip Fund	1.06; 1.06; 1.03
Low	ICICI Pru Liquid Fund; Kotak Liquid Fund; ABSL Liquid Fund	7.68; 6.18; 5.14
High	Kotak Flexicap Fund; ICICI Pru Value Discovery Fund; Kotak Emerging Equity Fund	0.98; 0.98; 0.96

The recommender is intentionally simple and transparent. It is an analytical demonstration rather than personalized financial advice. A production recommender would incorporate investor goals, horizon, liquidity needs, portfolio context, costs and suitability constraints.

13. Portfolio Concentration — HHI

The Herfindahl-Hirschman Index was calculated from portfolio holding weights. For each fund, the decimal holding weights were squared and summed, then scaled by 10,000.

Rank	Fund	HHI
1	Axis Bluechip Fund - Regular - Growth	2064.48
2	ABSL Small Cap Fund - Regular - Growth	2007.00
3	SBI Small Cap Fund - Direct Plan - Growth	1747.51
4	UTI Nifty 50 Index Fund - Regular - Growth	1747.09
5	Nippon India Large Cap Fund - Regular - Growth	1682.98

Higher HHI indicates greater concentration of portfolio weight among holdings. The analysis covered 34 funds with available holdings.

Axis Bluechip recorded the highest HHI in the analyzed holdings dataset at approximately 2064.48.

14. Interactive Power BI Dashboard

The Power BI dashboard converts the analytical outputs into an interactive reporting experience. The main pages are designed around business questions rather than individual datasets.

Page	Main content	Interaction
Industry Overview	AUM, SIP, folios, schemes, AUM trend, fund-house AUM	Industry/fund-house exploration
Fund Performance	Risk-return scatter, performance table, NAV trend	Fund house, category and plan slicers
Investor Analytics	State activity, transaction type, age and transaction volume	State, age group and city-tier slicers
SIP & Market Trends	SIP inflows, benchmark trend, category heatmap, top categories	Category/month exploration

A NAV drill-through page was also created for scheme-level investigation. The dashboard file is stored as `Mutual_Fund_Analytics_Bluestock.pbix`.

15. Dashboard Screenshots

The final presentation and report should include screenshots of the four main Power BI pages. The following layout is reserved for the final exported screenshots.

Industry Overview Insert Power BI screenshot here
Fund Performance Insert Power BI screenshot here
Investor Analytics Insert Power BI screenshot here
SIP & Market Trends Insert Power BI screenshot here

Replace these placeholders with the final dashboard screenshots before formal submission.

16. Limitations

- Historical analysis depends on the quality, completeness and period coverage of the supplied datasets.
- Historical VaR and CVaR describe observed return distributions and are not forecasts.
- Investor transaction data represents recorded activity and may not capture all real-world investor behavior.
- SIP continuity is based on a rule using observed transaction gaps and should not be interpreted as confirmed churn.
- The recommender uses a simplified risk mapping and Sharpe ranking rather than a full suitability model.
- Portfolio HHI is based on available holdings and may not represent the entire portfolio of every scheme.
- Power BI dashboard values are dependent on the loaded dataset and refresh state.
- Aggregate industry measures and scheme-level measures should not be mixed because they use different units and levels of aggregation.

17. Recommendations

17.1 Analytical recommendations

- Use risk-adjusted metrics alongside absolute returns when comparing schemes.
- Monitor VaR/CVaR and rolling Sharpe periodically rather than relying on one historical snapshot.
- Track SIP continuity as an early-warning indicator, while validating classifications with additional business rules.
- Use HHI and sector allocation to identify concentration risks.
- Extend cohort analysis with retention and tenure measures when longitudinal data becomes available.

17.2 Technical recommendations

- Automate weekday NAV ingestion and validation.
- Add a master `run_pipeline.py` script to execute the ETL workflow consistently.
- Introduce automated tests for row counts, duplicate keys, nulls and data types.
- Add scheduled Power BI refresh or a lightweight web application for recurring reporting.
- Maintain a clear separation between raw, processed and database outputs.

18. Conclusion and Deliverables

The BlueStock Mutual Fund Analytics capstone demonstrates a complete analytics lifecycle: raw financial data is ingested and cleaned, structured for SQL analysis, explored through visual analytics, evaluated using performance and risk metrics, extended with advanced investor and portfolio analysis, and presented through Power BI.

The project combines technical implementation with business interpretation. Key outputs include a 40-scheme risk analysis, investor and SIP behavior analysis, a transparent fund recommender, portfolio concentration metrics and an interactive dashboard.

Final deliverables

- Final_Report.pdf
- Bluestock_MF_Presentation.pptx
- Power BI dashboard (.pbix)
- EDA and performance notebooks
- Advanced analytics notebook
- ETL and supporting Python scripts
- SQL schema and queries
- Processed analytical CSV outputs
- README.md and project documentation

Final assessment: The project provides a reproducible foundation for mutual-fund analytics while clearly documenting assumptions, limitations and opportunities for further automation.

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